**Table 8.4** Reactions and Conversions in Second-Generation Bioethanol Production

| Reaction | Conversion |
|---|---|
| Glucose → 2 Ethanol + $2CO_2$ | Glucose 0.92 |
| Glucose + $1.2NH_3$ → 6 *Z. mobilis* + $2.4H_2O$ + $0.3O_2$ | Glucose 0.04 |
| Glucose + $2H_2O$ → Glycerol + $O_2$ | Glucose 0.002 |
| Glucose + $2CO_2$ → 2 Succinic acid + $O_2$ | Glucose 0.008 |
| Glucose → 3 Acetic acid | Glucose 0.022 |
| Glucose → 2 Lactic acid | Glucose 0.013 |
| 3 Xylose → 5 Ethanol + $5CO_2$ | Xylose 0.8 |
| Xylose + $NH_3$ → 5 *Z. mobilis* + $2H_2O$ + $0.25O_2$ | Xylose 0.03 |
| 3 Xylose + $5H_2O$ → 5 Glycerol + $2.5O_2$ | Xylose 0.02 |
| 3 Xylose + 5 $CO_2$ → 5 Succinic acid + $2.5O_2$ | Xylose 0.03 |
| 2 Xylose → 5 Acetic acid | Xylose 0.01 |
| 3 Xylose → 5 Lactic acid | Xylose 0.01 |

entrance of air into the system. Fermentation of pentoses is more complex. *Zymomonas mobilis* has been identified as appropriate for fermenting pentoses and hexoses simultaneously. In addition to ethanol, byproducts such as glycerol, succinic acid, acetic acid, and lactic acid are also produced. Table 8.4 shows the various reactions and the typical conversions in second-generation bioethanol processes (Aden and Foust, 2009). See Fig. 2.6 for a block diagram of the process.

The reactions that yield ethanol are exothermic:

$$C_6H_{12}O_6 \xrightarrow{\text{Yeast}} 2C_2H_5OH + 2CO_2 \qquad \Delta H = -84.394 \text{ kJ/mol}$$

$$3C_5H_{10}O_5 \xrightarrow{\text{Yeast}} 5C_2H_5OH + 5CO_2 \qquad \Delta H = -74.986 \text{ kJ/mol}_{xylose}$$

The reaction time is about 24 h at 0.12 MPa, which helps prevent the entrance of air into the system. The maximum concentration of ethanol in the water is 6–8%.

**EXAMPLE 8.1**

Model the fermentor for second-generation bioethanol production based on Krishnan et al. (1999).

**Solution**

The kinetic model for ethanol production via fermentation is based on the Michaelis–Menten mechanism:

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

In the case of ethanol production, there is both substrate and product inhibition:

$$\mathrm{EI} + \mathrm{S} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{E} + \mathrm{S} + \mathrm{I} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{ES} + \mathrm{I} \xrightarrow{k_2} \mathrm{E} + \mathrm{P} + \mathrm{I}$$

Assuming a steady-state for [E], [EI], and finally [ES], the modified Monod kinetics are given as follows:

$$\mu = \frac{\mu_m \cdot \mathrm{S}}{K_s + \mathrm{S} + \mathrm{S}^2/K_i}$$

Next, the product inhibition is included:

$$\frac{\mu}{\mu_0} = \left(1 - \left(\frac{\mathrm{P}}{\mathrm{P}_m}\right)^{\beta}\right)$$

The kinetics models for the different species involved are given in the equations that follow, where G represents glucose and X represents xylose. The parameters for the fermentation are given in Table 8E1.1 (Krishnan et al., 1999).

**Table 8E1.1** Kinetic Parameters for Fermentation

| Parameter | Glucose Fermentation | Xylose Fermentation |
|---|---|---|
| $\mu_m$ ($h^{-1}$) | 0.662 | 0.190 |
| $\nu_m$ ($h^{-1}$) | 2.005 | 0.250 |
| $K_s$ (g/L) | 0.565 | 3.400 |
| $K_s'$ (g/L) | 1.342 | 3.400 |
| $K_i$ (g/L) | 283.700 | 18.100 |
| $K_i'$ (g/L) | 4890.000 | 81.300 |
| $P_m$ (g/L) | 95.4 for $P \leq 95.4$ g/L | |
| | 129.9 for $95.4 \leq P \leq 129$ g/L | 59.040 |
| $P_m'$ (g/L) | 103 for $P \leq 103$ g/L | |
| | 136.4 for $103 \leq P \leq 136.4$ g/L | 60.200 |
| $\beta$ | 1.29 for $P \leq 95.4$ g/L | |
| | 0.25 for $95.4 \leq P \leq 129$ g/L | 1.036 |
| $\gamma$ | 1.42 for $P \leq 95.4$ g/L | 0.608 |
| $m$ ($h^{-1}$) | 0.097 | 0.067 |
| $Y_{P/S}$ (g/g) | 0.470 | 0.400 |
| $Y_{X/S}$ (g/g) | 0.115 | 0.162 |

Cells:

$$\mu_g = \frac{\mu_{m,g} \cdot \mathrm{S}}{K_{s,g} + \mathrm{S} + \mathrm{S}^2/K_{i,g}} \left(1 - \left(\frac{\mathrm{P}}{\mathrm{P}_m}\right)^{\beta_g}\right)$$

$$\mu_x = \frac{\mu_{m,x} \cdot \mathrm{S}}{K_{s,x} + \mathrm{S} + \mathrm{S}^2/K_{i,x}} \left(1 - \left(\frac{\mathrm{P}}{\mathrm{P}_m}\right)^{\beta_g}\right)$$

$$\frac{1}{\mathrm{X}}\frac{d\mathrm{X}}{dt} = \frac{\mathrm{G}}{G+X}\mu_g + \frac{\mathrm{X}}{G+X}\mu_x$$

Product:

$$\nu_{\mathrm{E},g} = \frac{v_{m,g} \cdot \mathrm{S}}{K_{s,g} + \mathrm{S} + \mathrm{S}^2/K_{i,g}} \left(1 - \left(\frac{\mathrm{P}}{\mathrm{P}_m}\right)^{\gamma_g}\right)$$

$$\nu_{\mathrm{E},x} = \frac{v_{m,x} \cdot \mathrm{S}}{K_{s,x} + \mathrm{S} + \mathrm{S}^2/K_{i,x}} \left(1 - \left(\frac{\mathrm{P}}{\mathrm{P}_m}\right)^{\gamma_x}\right)$$

with

$$\mu_{m,g} = 0.152 \cdot \mathrm{X}^{-0.461}$$
$$\mu_{m,x} = 0.075 \cdot \mathrm{X}^{-0.438}$$
$$v_{m,g} = 1.887 \cdot \mathrm{X}^{-0.434}$$
$$v_{m,x} = 0.16 \cdot \mathrm{X}^{-0.233}$$

$$\frac{1}{\mathrm{X}}\frac{d\mathrm{P}}{dt} = (\nu_{\mathrm{E},x} + \nu_{\mathrm{E},g})$$

Sustrate:

$$-\frac{d\mathrm{S}}{dt} = \frac{1}{Y_{\mathrm{X/S}}}\frac{d\mathrm{X}}{dt} + m\mathrm{X} = \frac{1}{Y_{\mathrm{P/S}}}\frac{d\mathrm{P}}{dt}$$

$$-\frac{d\mathrm{S}}{dt} = \frac{1}{Y_{\mathrm{P/S}}}\frac{d\mathrm{P}}{dt}$$

$$-\frac{d\mathrm{xylo}}{dt} = \frac{1}{Y_{\mathrm{P/S}}}(\nu_{\mathrm{E},x}\mathrm{X})$$

$$-\frac{d\mathrm{glu}}{dt} = \frac{1}{Y_{\mathrm{P/S}}}(\nu_{\mathrm{E},g}\mathrm{X})$$

The model for solving the equations can be written in any software. The following is example code for MATLAB:

```
[a,b]=ode15s('Fermen',[0 36],[200,50,0,1]);

plot(a,b(:,1),'k',a,b(:,2),'k--',a,b(:,3),'ko',a,b(:,4),'k.')
xlabel('t(h)')
ylabel('g/L')
legend('Glucose','Xylose','Ethanol','Cells')

function Reactor=Fermen(t,x)
```

```
Glucose = x(1);
xylose = x(2);
ethanol = x(3);
cells = x(4);
v_m_g = 2.005;
v_m_x = 0.250;
K_s_g = 0.565;
K_i_g = 283.7;
K_s_x = 3.4;
K_i_x = 18.1;
K_sp_g = 1.341;
K_ip_g = 4890;
K_sp_x = 3.4;
K_ip_x = 81.3;
m_g = 0.097;
m_x = 0.067;
Y_P_S_g = 0.470;
Y_P_S_x = 0.4;
P_m_x = 59.04;
P_mp_x = 60.2;
Beta_x = 1.036;
gamma_x = 0.608;
if ethanol <95.4;
P_m_g = 95.4;
Beta_g = 1.29;
gamma_g = 1.42;
else
P_m_g = 129;
Beta_g = 0.25;
gamma_g = 0;
end
if ethanol <103;
P_mp_g = 103;
else
P_mp_g = 136;
end

if cells < 5;
mu_m_g = 0.152*(cells)^(-0.461);
v_m_g = 1.887*(cells)^(-0.434);
mu_m_x = 0.075*(cells)^(-0.438);
v_m_x = 0.16*(cells)^(-0.233);
else
```

```
mu_m_g =0.662;
v_m_g= 2.005;
mu_m_x=0.190;
v_m_x= 0.25;
end
mu_g=mu_m_g*Glucose*(1-(ethanol/P_m_g)^Beta_g)/
  (K_s_g+Glucose+Glucose^2/K_i_g);
mu_x=mu_m_x*xylose*(1-(ethanol/P_m_x)^Beta_x)/
  (K_s_x+xylose+xylose^2/K_i_x);
v_g=v_m_g*Glucose*(1-(ethanol/P_mp_g)^gamma_g)/
  (K_sp_g+Glucose+Glucose^2/K_ip_g);
v_x=v_m_x*xylose*(1-(ethanol/P_mp_x)^gamma_x)/
  (K_sp_x+Glucose+Glucose^2/K_ip_x);
Reactor(1,1)=-(1/Y_P_S_g)*v_g*cells;
Reactor(2,1)=-(1/Y_P_S_x)*v_x*cells;
Reactor(3,1)=(v_g+v_x)*cells;
Reactor(4,1)=cells*((Glucose)*mu_g/(Glucose+xylose)+
  (xylose)*mu_x/(Glucose+xylose));
```

The profile of the species over time can be seen in (Fig. 8E1.1)

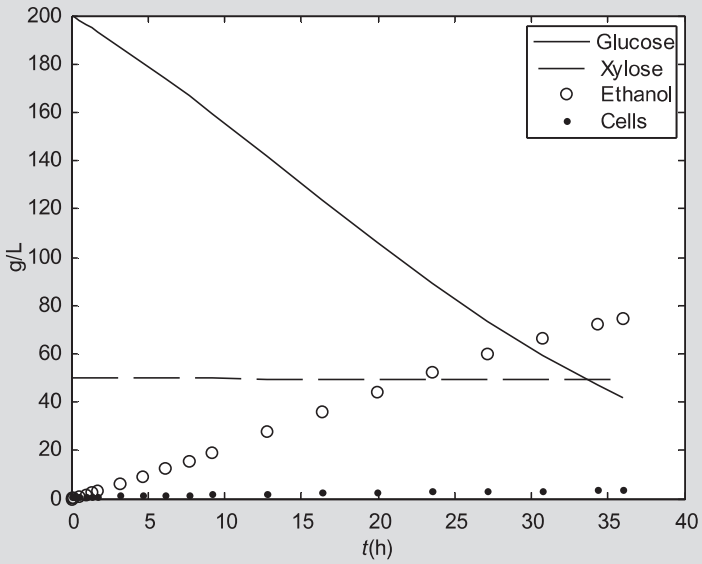



**FIGURE 8E1.1**

Ethanol production from hexoses and pentoses.

**Table 8.5** Conversions of Sugars into FAEE and Byproducts

| Reaction | Conversion | $DH_r$ (kJ/mol) |
|---|---|---|
| 9 Glucose + $2O_2 \rightarrow 2C_{18}H_{36}O_2 + 18CO_2 + 18H_2O$ | Glucose 0.3 | −405 |
| Glucose → 2 Ethanol + $2CO_2$ | Glucose 0.3 | −84,394 |
| Glucose + $1.2NH_3$ → 6 *Z. mobilis* + $2.4H_2O + 0.3O_2$ | Glucose 0.1 | NA |
| Glucose + $2H_2O \rightarrow 2$ Glycerol + $O_2$ | Glucose 0.3 | 504 |
| 27 Xylose + $5O_2 \rightarrow 5C_{18}H_{36}O_2 + 45CO_2 + 45H_2O$ | Xylose 0.2 | −338 |
| 3 Xylose → 5 Ethanol + $5CO_2$ | Xylose 0.2 | −74,986 |
| Xylose + $NH_3$ → 5 *Z. mobilis* + $2H_2O + 0.25O_2$ | Xylose 0.2 | NA |
| 3 Xylose + $5H_2O \rightarrow 5$ Glycerol + $2.5O_2$ | Xylose 0.2 | 418 |

### 8.2.1.2 Biodiesel

Glucose and xylose have recently been fermented into biodiesel (fatty acid ethyl esters, FAEE). The fermentation takes place aerobically at 32–38°C, with a yield of 0.35 g/g sugar. The concentration of FAEE in the reactor is fixed at 0.01 kg/kg. Biodiesel is immiscible in water, and therefore the separation of the organic and aqueous phases is easy. However, ethanol and glycerol are also produced, and remain in the aqueous phase. This reduces the yield of carbon to FAEE. Table 8.5 shows some typical conversions (Martín and Grossmann, 2015).

### 8.2.1.3 Ibutene

Sugars can also be a source of valuable chemicals. Ibutene, a C4 chemical, can be produced from glucose fermentation. The advantage of this process is that the product is in the gas phase and thus no energy-intensive dehydration is required. Ibutene must be separated from the $CO_2$ (Martín and Grossmann, 2014).

$$C_6H_{12}O_6 \xrightarrow{\text{Yeast}} C_4H_8 + 2CO_2 + 2H_2O + 2\ \text{ATP} \qquad \Delta H = -41.9\ \text{kJ/mol}$$

### 8.2.1.4 Furans

Dehydration of C6 sugars yields hydroxymethyl furfural, an intermediate in the production of dimethyl furfural (DMF). C5 sugars can produce furfural as presented in the chemical reactions below:

